Real-World Scale Drawings

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lesson 4 — Real-World Scale Drawings	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.GR.1.5

2. Standards

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Must. Read a scale such as ¼ in : 1 ft or 1 : 50 as a rate, find real lengths and drawing lengths from it in both directions, and find a real area from the real dimensions. (MA.7.GR.1.5, clarifications 1 and 3; B1G-M's wall example.)

Must not. Converting the real area into another system (5.08–09); drawing a plan to scale with a ruler (not a FAST shape — the audit's ruler items are out); a scale factor between two drawings of one object (5.01).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.1.1	Actively participate in effortful learning both individually and collectively.	Example 2 and its Your Turn: the scale is not printed, and the room has to make it from one pair before the question can start
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	every board is a plan, a map or a model: the scale is a real object's, and the answer is checked against what a room or a car could be

4. Learning Target and Essential Question

Learning target. I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

Essential question. A room is 2 in by 3 in on a plan at 1 in : 4 ft. Why is its real area not 6 × 4 square feet?

Building on. 5.01 — a map scale used both ways; 5.02–03 — area is two lengths, so the rate is used twice; Grade 6 — a ratio written as a unit rate.

Working toward. 5.06–07 — a conversion ratio is a rate too, and it is multiplied the same way; 5.10–11 — a scale, then a speed: two rates in one problem.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A plan's scale is ¼ in : 1 ft. Why does writing it as 1 in : 4 ft make every question easier?	DOK 2
A room is 2 in by 3 in on that plan. Why is its real area 96 ft^2 and not 24?	DOK 3
Two models of one plane are built at 1 : 48 and 1 : 72. Which is longer, and why does the bigger number make the smaller model?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A plan is drawn at $\frac{1}{2}\text{ in} : 1\text{ ft}$. A wall is 4 in long on it. How long is the real wall?	***8 ft***	2 — divided (4 × ½) [Notes I: per 1 inch first, then × 2]; 4 — read the scale as 1 : 1
2	A plan is drawn at $1\text{ in} : 3\text{ ft}$. A room is 2.5 in long on it. How long is the real room?	***7.5 ft***	5.5 — added 3 [7.GR.1.5: the additive error — Math Nation's named misconception]; 0.83 — divided [Notes I: drawing to real is × rate]
3	A plan is drawn at $1\text{ in} : 3\text{ ft}$. A real wall is 21 ft long. How long is it on the plan?	***7 in***	63 — multiplied [Notes III: real to drawing is a division]; 18 — subtracted 3
4	A model car is built at $1 : 50$ and is 8 cm long. How long is the real car?	***A — 4 m***	(B) 8 ÷ 50: divided [Notes I: drawing to real is × rate]; (C) 400 with the wrong unit: 400 cm, not 400 m [Notes I: the scale's units are the drawing's]; (D) 8 + 50: the additive error [B1G-M 7.GR.1.5]
5	A plan is drawn at $\frac{1}{4}\text{ in} : 1\text{ ft}$. A real wall is 15 ft long. How long is it on the plan?	***3.75 in***	60 — multiplied by 4 [Notes III: real to drawing is ÷]; 7.5 — divided by 2
6	A plan is drawn at $1\text{ in} : 4\text{ ft}$. A room is 2 in by 3 in on it. What is the room's real area?	***B — 96 ft²***	(A) 6 in² × 4: the drawing's area scaled by the length rate [B1G-M 7.GR.1.5: scaling area like length; Notes II]; (C) 2(8 + 12): the real perimeter, not the area [5.02–03 Notes I]; (D) 8 + 12: the real dimensions added [7.GR.1.5: the additive error — Math Nation's named misconception]
7	A plane is 36 ft long. One model of it is built at $1 : 48$, another at $1 : 72$. Which model is longer, and by how much?	***The 1 : 48 model, by 3 in (9 in against 6 in)***	the 1 : 72 model — 'the bigger scale is the bigger model' [Notes III: real ÷ rate — a bigger rate divides more]; 0.75 ft and 0.5 ft with no comparison — stopped early
8	Mia, who is 13, enlarges her 9 in by 12 in poster into a mural. The mural's scale is $1\text{ in} : 2\text{ ft}$. What are the mural's dimensions?	***18 ft by 24 ft***	4.5 ft by 6 ft — divided [Notes I: drawing to real is × rate]; 18 in by 24 in — kept the drawing's unit
9	On Andrew's drawing a wall is 3.5 in wide and 4 in tall; the real wall is 26.25 ft wide. Find the real height and the real area, and say which step uses the scale.	***30 ft high; 787.5 ft². The scale (7.5 ft per in) is the step from 4 in to 30 ft.***	105 ft² — 3.5 × 4 = 14 in², then × 7.5 [B1G-M 7.GR.1.5: scaling area like length]; 'the area step uses the scale' — the scale is between a drawing and a real length, not two real ones

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Every scale is rewritten per 1 drawing unit on the board before a question is read, and stays there. Two lines under it all period: drawing × rate = real; real ÷ rate = drawing.

ELL. **Scale** here is a ratio, not the kitchen scale; **plan** means a drawing from above. Say "on the paper" for drawing lengths and "in the room" for real ones with every board.

Enrichment. Board 7 finishes early for some: ask them to write the two models' lengths in centimeters with 1 in = 2.54 cm — Monday's lesson, a day early.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the scale made from the pair (7.5 ft per inch, as Example 2 did), the height from it (30 ft), the area from the two real dimensions (787.5 ft^2), and a sentence naming which step uses the scale. A board that scales $3.5 \times 4 = 14$ in^2 by 7.5 has not gone to real units first.

Vocabulary. **floor plan**; **scale (two units)**; **unit rate of a scale**; **scale with one unit**

Materials. Whiteboards and markers. A printed floor plan under the document camera for Notes II if the slide's rectangle is not enough. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Scale drawings: word problems — 84H; Scale drawings: scale factor word problems — KCM.